Docker Images

Docker File

A Dockerfile is a text file used to define the instructions for building a Docker image. It provides a declarative and reproducible way to create an image with all the necessary components and configurations needed to run an application.

The Dockerfile consists of a series of instructions that are executed in order to create the image layer by layer. Each instruction represents a step in the build process, such as installing dependencies, copying files, setting environment variables, and running commands.

Docker Image

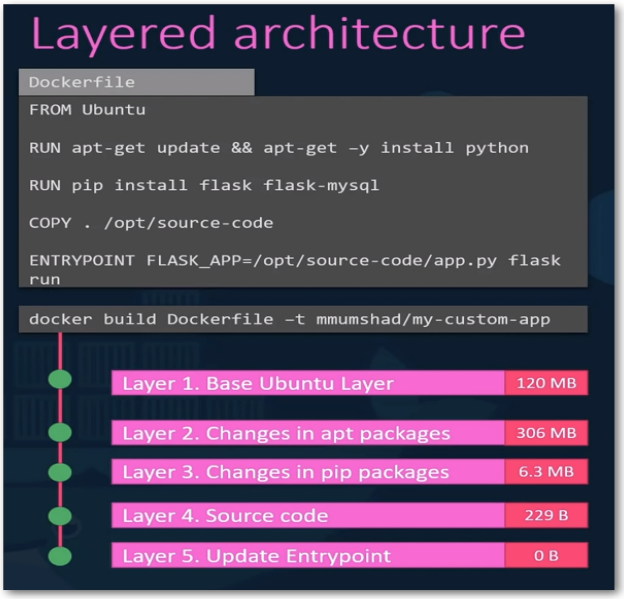
A Docker image is a lightweight, standalone, and executable software package that contains everything needed to run a piece of software, including the code, runtime environment, libraries, and system tools. It serves as a template for creating Docker containers, which are the running instances of Docker images.

Docker images are built from a set of instructions defined in a text file called a Dockerfile. The Dockerfile specifies the base image to start from, the dependencies to install, any configurations to be made, and the commands to run when the image is launched.

Docker Build :

The docker build command is used to build a Docker image from a Dockerfile. It reads the instructions defined in the Dockerfile and creates a new image based on those instructions

Docker Image Layers



Docker images follow the layered architecture, where each instruction in the Dockerfile adds a new layer to the image. This layered approach allows for efficient storage and sharing of images. If changes are made to an image, only the affected layers need to be updated, while the rest can be reused from the cached layers